

$$\begin{aligned}
1. \text{ A) } L(y|\theta_0) &= \frac{1}{\theta_0^n} \exp\left(\frac{-\sum_{i=1}^n y_i}{\theta_0}\right) \\
L(y|\theta_a) &= \frac{1}{\theta_a^n} \exp\left(\frac{-\sum_{i=1}^n y_i}{\theta_a}\right) \\
\frac{L(y|\theta_0)}{L(y|\theta_a)} &= \frac{\theta_a^n}{\theta_0^n} \exp\left[\frac{-\sum_{i=1}^n y_i}{\theta_0} - \frac{-\sum_{i=1}^n y_i}{\theta_a}\right] < k \\
\exp\left[\frac{-\sum_{i=1}^n y_i}{\theta_0} - \frac{-\sum_{i=1}^n y_i}{\theta_a}\right] &< \frac{\theta_0^n}{\theta_a^n} k \\
\left[\left(\frac{1}{\theta_0} - \frac{1}{\theta_a}\right)(-\sum_{i=1}^n y_i)\right] &< n \ln \frac{\theta_0}{\theta_a} + \ln k \\
[-\sum_{i=1}^n y_i] &< (n \ln \frac{\theta_0}{\theta_a} + \ln k) \left(\frac{1}{\theta_0} - \frac{1}{\theta_a}\right)^{-1} \\
\sum_{i=1}^n y_i &> -(n \ln \frac{\theta_0}{\theta_a} + \ln k) \left(\frac{1}{\theta_0} - \frac{1}{\theta_a}\right)^{-1} \\
k' &= -(n \ln \frac{\theta_0}{\theta_a} + \ln k) \left(\frac{1}{\theta_0} - \frac{1}{\theta_a}\right)^{-1} \\
\sum_{i=1}^n y_i &> k'
\end{aligned}$$

B) The rejection region does not depend on θ_a

$$\begin{aligned}
\text{C) } i) \Omega_0 &= \{(\theta): \theta = 2\} & \Omega_a &= \{(\theta): \theta \neq 2\} \\
\max_{\theta \in \Omega_0} L(y|\theta) &= \frac{1}{\theta^n} \exp\left(\frac{-\sum_{i=1}^n y_i}{\theta}\right) = \frac{1}{2^n} \exp\left(\frac{-\sum_{i=1}^n y_i}{2}\right) \\
\max_{\theta \in \Omega_0 \cup \Omega_a} L(y|\theta) &= \frac{1}{\theta^n} \exp\left(\frac{-\sum_{i=1}^n y_i}{\theta}\right)
\end{aligned}$$

$$\text{Take log of } L(y|\theta) \rightarrow l(y|\theta) = -n \ln \theta + \frac{-\sum_{i=1}^n y_i}{\theta}$$

$$\frac{\partial l(y|\theta)}{\partial \theta} l(y|\theta) = \frac{-n}{\theta} + \frac{\sum_{i=1}^n y_i}{\theta^2}$$

$$\text{Set equal to 0} \rightarrow \hat{\theta} = \frac{\sum_{i=1}^n y_i}{n} = \bar{y}$$

$$\text{Plug } \bar{y} \text{ back into } \max_{\theta \in \Omega_0 \cup \Omega_a} L(y|\theta) \rightarrow \max_{\theta \in \Omega_0 \cup \Omega_a} L(y|\theta) = \frac{1}{\bar{y}^n} \exp(-n)$$

$$\lambda = \frac{\max_{\theta \in \Omega_0} L(y|\theta)}{\max_{\theta \in \Omega_0 \cup \Omega_a} L(y|\theta)} = \frac{\frac{1}{2^n} \exp\left(\frac{-\sum_{i=1}^n y_i}{2}\right)}{\frac{1}{\bar{y}^n} \exp(-n)} = \left(\frac{\bar{y}}{2}\right)^n \exp\left(\frac{-\sum_{i=1}^n y_i}{2} + n\right)$$

$$\text{ii) Plugging in } \sum_{i=1}^n y_i = 170 \text{ and } n=100 \rightarrow \lambda = \left(\frac{1.7}{2}\right)^{100} \exp\left(\frac{-170}{2} + 100\right) = 0.285963$$

Using Wilks' theorem $TS = -2 \log \lambda = 2.503786$ follows an approximate chi-squared distribution with 1 df

With a p-value of 0.113573, we fail to reject the null hypothesis. There is not enough evidence to conclude $\theta \neq 2$

iii) If a 95% confidence interval was constructed for θ , $\theta = 2$ would be in the interval since it is a plausible value.

iv) A LRT CI would be constructed by plugging possible values of θ into the test statistic and seeing if it is within the rejection region at the desired α level. Then all plausible values would be combined to form the confidence interval.

$$2. \Omega_0 = \{(\theta_1, \theta_2): \theta_1 = \theta_2 = \theta\}$$

$$L(x, y|\theta_1, \theta_2) = \frac{1}{\theta_1^{n_1}} * \frac{1}{\theta_2^{n_2}} \exp\left(\frac{-\sum_{i=1}^{n_1} x_i}{\theta_1} + \frac{-\sum_{i=1}^{n_2} y_i}{\theta_2}\right)$$

$$\max_{\theta \in \Omega_0} L(x, y|\theta) = \frac{1}{\theta^{n_1+n_2}} \exp\left(\frac{-\sum_{i=1}^{n_1} x_i - \sum_{i=1}^{n_2} y_i}{\theta}\right)$$

$$\text{Take log of } \max_{\theta \in \Omega_0} L(x, y|\theta) \rightarrow l(x, y|\theta) = -(n_1 + n_2) \log \theta +$$

$$\left(\frac{-\sum_{i=1}^{n_1} x_i - \sum_{i=1}^{n_2} y_i}{\theta}\right)$$

$$\frac{\partial l(\theta)}{\partial \theta} = \frac{-(n_1+n_2)}{\theta} + \frac{\sum_{i=1}^{n_1} x_i + \sum_{i=1}^{n_2} y_i}{\theta^2}$$

$$\text{Set equal to 0} \rightarrow \hat{\theta} = \frac{\sum_{i=1}^{n_1} x_i + \sum_{i=1}^{n_2} y_i}{n_1+n_2}$$

$$\Omega_0 \cup \Omega_a = \{(\theta_1, \theta_2): \theta_1 > 0, \theta_2 > 0\}$$

$$\max_{\theta \in \Omega_0 \cup \Omega_a} L(x, y|\theta_1, \theta_2) = \frac{1}{\theta_1^{n_1}} * \frac{1}{\theta_2^{n_2}} \exp\left(\frac{-\sum_{i=1}^{n_1} x_i}{\theta_1} + \frac{-\sum_{i=1}^{n_2} y_i}{\theta_2}\right)$$

$$\text{Take log of } \max_{\theta \in \Omega_0 \cup \Omega_a} L(x, y|\theta_1, \theta_2) \rightarrow l(x, y|\theta_1, \theta_2) = -n_1 \log \theta_1 -$$

$$n_2 \log \theta_2 + \left(\frac{-\sum_{i=1}^{n_1} x_i}{\theta_1} + \frac{-\sum_{i=1}^{n_2} y_i}{\theta_2}\right)$$

$$\frac{\partial l(x, y|\theta_1, \theta_2)}{\partial \theta_1} = \frac{-n_1}{\theta_1} + \frac{\sum_{i=1}^{n_1} x_i}{\theta_1^2}$$

$$\text{Set equal to 0} \rightarrow \widehat{\theta}_1 = \frac{\sum_{i=1}^{n_1} x_i}{n_1} = \bar{x}$$

$$\frac{\partial l(x, y|\theta_1, \theta_2)}{\partial \theta_2} = \frac{-n_2}{\theta_2} + \frac{\sum_{i=1}^{n_2} y_i}{\theta_2^2}$$

$$\text{Set equal to 0} \rightarrow \widehat{\theta}_2 = \frac{\sum_{i=1}^{n_2} y_i}{n_2} = \bar{y}$$

$$\lambda = \frac{\max_{\theta \in \Omega_0} L(x, y|\theta)}{\max_{\theta \in \Omega_0 \cup \Omega_a} L(x, y|\theta_1, \theta_2)} = \frac{\frac{1}{\widehat{\theta}^{n_1+n_2}} \exp\left(\frac{-\sum_{i=1}^{n_1} x_i - \sum_{i=1}^{n_2} y_i}{\widehat{\theta}}\right)}{\frac{1}{\bar{x}^{n_1}} * \frac{1}{\bar{y}^{n_2}} \exp\left(\frac{-\sum_{i=1}^{n_1} x_i}{\bar{x}} + \frac{-\sum_{i=1}^{n_2} y_i}{\bar{y}}\right)} =$$

$$\frac{\bar{x}^{n_1} * \bar{y}^{n_2} \exp\left(\frac{-\sum_{i=1}^{n_1} x_i - \sum_{i=1}^{n_2} y_i}{\widehat{\theta}}\right)}{\exp(-n_1 - n_2)} = \frac{\bar{x}^{n_1} * \bar{y}^{n_2}}{\widehat{\theta}^{n_1+n_2}} \exp\left(\frac{-\sum_{i=1}^{n_1} x_i - \sum_{i=1}^{n_2} y_i}{\widehat{\theta}} + n_1 + n_2\right)$$

$$3. A) \beta\left(\frac{\sigma_1^2}{\sigma_2^2}\right) = P\left(\frac{s_1^2}{s_2^2} \leq \Delta_0(F_{n_1-1, n_2-1; 1-\alpha})\right)$$

$$\beta\left(\frac{\sigma_1^2}{\sigma_2^2}\right) = P\left(\frac{s_1^2/s_2^2}{\Delta_0} \leq (F_{n_1-1, n_2-1; 1-\alpha})\right)$$

$$\beta\left(\frac{\sigma_1^2}{\sigma_2^2}\right) = P(F \leq (F_{n_1-1, n_2-1; 1-\alpha}))$$

$$B) \beta(2.5) = P(F \leq (F_{25-1, 33-1; 1-0.05}))$$

$$\beta(2.5) = P(F \leq F_{24, 32; 0.95})$$

```
pf(2.5, 24, 32, lower.tail=TRUE)
```

```
## [1] 0.9919608
```

C) To prevent from solving for multiple unknown variables, we assume $n_1 = n_2 = n$. Finding the minimum sample size can be found in R by using a function that runs a loop testing values of n and the corresponding type II error probabilities.

4. The code below sets up the vectors of cornflake and oatbran serum cholesterol levels and runs a paired t-test of the data.

$$H_0: \mu_c - \mu_o = \mu_d \leq 0 \quad H_a: \mu_c - \mu_o = \mu_d > 0$$

```
cornflk <- c(4.61, 6.42, 5.40, 4.54, 3.98, 3.82, 5.01, 4.34, 3.80, 4.56, 5.35, 3.89, 2.25, 4.24)
```

```
oatbran <- c(3.84, 5.57, 5.85, 4.80, 3.68, 2.96, 4.41, 3.72, 3.49, 3.84, 5.26, 3.73, 1.84, 4.14)
```

```
t.test(cornflk, oatbran, alternative="greater", mu=0, paired=TRUE, conf.level=0.05)
```

```
##
```

```
## Paired t-test
```

```
##
```

```
## data: cornflk and oatbran
```

```
## t = 3.3444, df = 13, p-value = 0.002639
```

```
## alternative hypothesis: true difference in means is greater than 0
```

```
## 5 percent confidence interval:
```

```
## 0.5550006 Inf
```

```
## sample estimates:
```

```
## mean of the differences
```

```
## 0.3628571
```

The test statistic is 3.3444 and follows a t-distribution with 13 df. With a low p-value of 0.002639, we reject the null hypothesis. There is strong evidence to conclude that eating oat bran lowers serum cholesterol levels. The lower limit of a 95% confidence interval is 0.5550006. Since 0 is not in the interval $(0.5550006, \infty)$, we would reject the null hypothesis here as well if we only calculated a confidence interval.